LOVENTA ANYANGO OTIENO

Graduate Geospatial Engineer
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PROFESSIONAL SUMMARY

Graduate Geospatial Engineer with a B.Sc. in Geomatic Engineering and Geospatial Information Systems (JKUAT, 2025) and hands-on experience in Geographic Information Systems (GIS), remote sensing, Global Navigation Satellite System (GNSS) surveying, total station surveying and land surveying. Experienced in engineering and topographic surveys, survey control establishment, cadastral surveys, aerial imagery digitisation, mapping and technical report writing for infrastructure, Affordable Housing Programme, river corridor, water and sanitation, and agricultural monitoring projects. Applies Python-based geospatial analysis, machine learning and deep learning to real-world environmental and infrastructure problems. Published co-author of two peer-reviewed journal articles on remote sensing and machine learning for forest ecosystem monitoring, and a book chapter on the effects of climate change on crop production. Skilled at translating field survey and satellite data into actionable maps, datasets and client-ready reports.

CORE SKILLS

- **GIS & Remote Sensing:** ArcGIS, QGIS, Google Earth Engine, ERDAS Imagine, LiDAR, Aerial Imagery Digitisation, Satellite Image Processing, Mapping
- **Programming & Data Analysis:** Python, R, JavaScript, PostgreSQL, PostGIS, Machine Learning, Deep Learning
- **Surveying & CAD:** AutoCAD, Civil 3D, Total Station, RTK (Real-Time Kinematic) GNSS, Static GNSS, Topographic Surveying, Engineering Surveys, Cadastral Surveying
- **Cloud & Data Platforms:** Oracle Cloud Infrastructure, Copernicus, Earthdata, USGS EarthExplorer
- **Core Competencies:** Technical Report Writing, Communication, Critical Thinking, Research, Teamwork, Problem-Solving

PROFESSIONAL EXPERIENCE

Graduate Geospatial Engineer, Nile Surveys & Geo-Solutions Limited, Kenya *Dec 2025 – Present*

- Processed and digitised aerial imagery to generate spatial datasets for mapping and analysis.
- Prepared location, topographic and thematic maps for various engineering, infrastructure and land development projects.
- Conducted GNSS static observations, survey control establishment and beacon re-establishment using RTK GNSS and Total Station equipment.
- Undertook engineering and topographic surveys for road corridors and other infrastructure projects.
- Compiled field observations, survey results and mapping outputs into technical reports and project deliverables.

Earth Observation Intern, Kenya Space Agency, Kenya *Jan 2024 – Apr 2024*

- Developed Python scripts to process satellite imagery and visualise remote sensing outputs, for agricultural monitoring under the Agriculture Information Management System (AIMS) for Murang'a County.
- Trained Microsoft AI through data annotation for the AIMS project.
- Reviewed and edited Space Club educational content to improve scientific accuracy and readability for young audiences.
- Served as rapporteur for the 'Development of Earth Observation decision support tools for the agricultural sector' workshop, hosted by KSA, compiling and presenting a comprehensive final report.
- Coordinated data entry and logistics for Kenya Space Expo 2024, supporting smooth event delivery.

Intern, Ministry of Lands, Public Works, Housing and Urban Planning, Kenya *Jan 2023 – Mar 2023*

- Conducted general boundary surveys using RTK GNSS and hand-held GPS under supervision and prepared mutation forms for clients.
- Conducted cadastral surveys using RTK GNSS under supervision.
- Verified and organised land information to improve data accuracy, accessibility and consistency.
- Performed additional administrative and data management duties assigned by supervisors.

SELECTED PROJECT EXPERIENCE

Consultancy Services for the Development of Bankable Water and Sanitation Proposals for Water and Sanitation Service Providers

Report Writing / GIS Assistant | Client: Government of Kenya | Kenya | Sep 2026

- Compiled and organised collected GIS, survey and spatial datasets for use in the project reports.
- Reviewed and processed spatial data to support preparation of maps, figures and technical outputs.
- Prepared and edited technical reports, incorporating field findings, processed datasets and project deliverables.
- Prepared maps and spatial illustrations for inclusion in the technical reports.
- Reviewed datasets and deliverables for completeness, consistency and presentation quality before submission.

- Consolidated inputs from different technical activities into structured client-ready reports and deliverables.
- Supported the preparation and formatting of final documentation for submission to the client.

Tavevo Waters – Aerial Mapping and Land Survey

Report Writing / GIS Assistant | Client: Tavevo Waters | Kenya | Jul 2026 – Aug 2026

- Prepared site location and index maps for Tavevo Waters project sites for inclusion in the technical report.
- Compiled and organised survey, aerial mapping and processed spatial datasets for reporting.
- Used QGIS to digitise and edit mapped features from aerial imagery and survey data.
- Prepared and incorporated GIS maps, figures and spatial data outputs into the technical report.
- Reviewed processed datasets and mapping outputs for completeness and consistency before finalisation.
- Compiled the survey findings, maps and processed datasets into a client-ready technical report for submission.
- Assisted in the formatting, presentation and quality review of final report deliverables.

Affordable Housing Programme (AHP) – Mapping for Selected Sites in Kenya (Eco-plan Ltd)

Assistant Surveyor | Clients: Eco Plan Kenya Ltd and MS. Renassance Planning Ltd | Kenya | Feb 2026 – Apr 2026

- Conducted static GNSS surveys at selected Affordable Housing Programme (AHP) sites to establish and verify survey control.
- Processed and reviewed GNSS observations and prepared control survey records for the surveyed sites.
- Documented control point coordinates, physical descriptions and site locations in survey reports.
- Used QGIS to digitise existing site features, including access roads, drainage networks, buildings and other infrastructure.
- Prepared site location and vicinity maps for inclusion in the AHP technical reports.
- Compiled survey and GIS data into spatial datasets and basemaps for site planning and project development.
- Assisted in the preparation and quality review of survey drawings, maps and technical deliverables.
- Supported the survey team in field data collection, control verification and site mapping activities.

Mapping for Nairobi River Protection, Restoration, Reconstruction & Beautification Project (Phase II)

Assistant Surveyor | Client: CEEC Guanxi Electric Power (Kenya) Co. Ltd | Nairobi, Kenya | Dec 2025 – Jul 2026

- Executed topographic field surveys along selected sections of the Nairobi River corridor using Total Station and RTK GNSS equipment.
- Captured ground levels, road alignments, river channel details, drainage features and adjacent infrastructure to support project mapping.
- Undertook detail surveying of selected road corridors and surrounding areas to establish existing site conditions.
- Transferred and organised field survey observations for preparation of digital mapping outputs.
- Applied QGIS for spatial data digitisation and editing, converting surveyed features into structured GIS datasets.
- Prepared spatial information illustrating river corridor features, terrain characteristics and existing infrastructure for project planning and analysis.
- Assisted in reviewing survey data and GIS outputs to ensure consistency between field observations and mapped features.

Evaluating the Role of Land Surface Temperature in Post-Fire Vegetation Recovery: A Case Study of Mt. Kenya (Final Year Research Project)

Researcher | Institution: JKUAT (Supervisor: Dr. Harison Kipkulei)| Kenya | 2024-2025

- Mapped burn severity of the 2012 Mt. Kenya fire and tracked vegetation recovery through 2021 using Landsat 7/8 time series (NBR/dNBR) processed in Google Earth Engine and QGIS.
- Integrated Land Surface Temperature, climate, soil organic carbon and SRTM terrain variables into a Python Random Forest regression model to predict recovery (test R^2 0.90, RMSE 0.028; 10-fold cross-validated R^2 0.88).
- Applied ANOVA and Tukey HSD to test recovery differences across burn severity classes, and used feature importance and partial dependence analysis to interpret drivers.
- Cross-checked the mapped burned area against Kenya Forest Service records and the MODIS MCD64A1 product.
- Findings published in Discover Geoscience (2025).

EDUCATION

B.Sc. Geomatic Engineering and Geospatial Information Systems, Jomo Kenyatta University of Agriculture and Technology (JKUAT), Juja *Sep 2019 – June 2025*

Second Class Honours (Lower Division)

Kenya Certificate of Secondary Education (KCSE), Moi Forces Academy, Lanet *Feb 2015 – Nov 2018*

Grade A (82 points)

PUBLICATIONS

- Kipkulei, H., Awandu, W., Otieno, T. A., Otieno, L. A., Ngoya, Z. J., & Rotich, B. (2027). Consequences of climate change on crop production. In T. M. Letcher (Ed.), Climate change (pp. 239–259). Elsevier. https://doi.org/10.1016/B978-0-443-32834-3.00010-4
- Otieno, L. A., Otieno, T. A., Rotich, B., Löhr, K., & Kipkulei, H. K. (2025). Integrating remote sensing and machine learning to evaluate environmental drivers of post-fire vegetation recovery in the Mount Kenya forest. Discover Geoscience, 3(1), 81. https://doi.org/10.1007/s44288-025-00196-5
- Otieno, T. A., Otieno, L. A., Rotich, B., Löhr, K., & Kipkulei, H. K. (2025). Modeling climate change impacts and predicting future vulnerability in the Mount Kenya forest ecosystem using remote sensing and machine learning. Environmental Monitoring and Assessment, 197(6), 631. https://doi.org/10.1007/s10661-025-14089-0

CERTIFICATIONS

- Python for Data Science, AI & Development – IBM (Sep 2026)
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle (Aug 2025 – Aug 2027)

PROFESSIONAL MEMBERSHIP

Graduate Member, Institution of Surveyors of Kenya (ISK) – Membership No. 7081

LANGUAGES

English (Fluent) | Swahili (Fluent) | Luo (Fluent) | French (Intermediate)

REFEREES

Dr. rer. agr. Harison Kipkulei – Lecturer, Department of Geomatic Engineering, JKUAT | hkipkulei@jkuat.ac.ke | 0717 623 982
Mr. Conrad Olimba– Surveyor, Nile Surveys & Geo-Solutions Limited | 0703 305 553